

Basic Prob Theory + Related Results:

Probability: Mathematical study of a random phenomenon or random experiment.

Experiment $\begin{cases} \text{Deterministic Exp} \\ \text{Undeterministic or Random exp.} \end{cases}$

Deterministic Exp: Sure about the outcome, Ex. $V = u + at$

Random Exp: Not sure about the outcome only various possibilities are known

Ex: Toss a coin, throw a dice, Temp at a place well in advance, No. of defects in per 100 meter of cloths, Profit of a firm in advance etc.

Sample Space: Set of all possible outcomes in a random experiment. denoted by S or Ω

- Ex:
- i) Toss a coin $S = \{H, T\}$
 - ii) Toss two coins $S = \{HH, HT, TH, TT\}$
 - iii) Throw a dice $S = \{1, 2, 3, 4, 5, 6\}$
 - iv) Throw two dice $S = \{(1,1), (1,2) \dots (6,6)\}$
 - v) Temp of a place $S = \{T: T_1 \leq T \leq T_2\}$
 - vi) Profit of a firm $S = \{P: -\infty < P < \infty\}$
 - vii) No. of defects $S = \{D: D = 1, 2, 3, \dots\}$

Event: Any subset of sample space in which one is interested, denoted by $A, B, C \dots$ etc.

Ex: $A = \{H\}$, $B = \{T\}$, $C = \{\}$, $D = \{H, T\}$
 C is called an impossible event denoted by ϕ
 D is called a sure event that is whole sample space.

σ -field : denoted by $\mathcal{A}, \mathcal{B}, \mathcal{C}$ etc.

Set of all possible subsets of a sample space.

Ex: RE: Toss a coin, $S = \{H, T\}$

$$\mathcal{A} = \{ \{H\}, \{T\}, \emptyset, S \}$$

Terms + types of events

- I) Equally likely events
- II) Exhaustive events
- III) Mutually Exclusive / Mutually disjoint events

Various approaches of Probability

I) Classical or Mathematical Approach:

If a random experiment results in n (finite) ~~max~~ m excl. and equally likely cases and out of such cases if m no. of cases favour an event A , then the

$$P(A) = \frac{m}{n} \quad \text{it is obvious that } P(\emptyset) = \frac{0}{n} = 0$$

$$P(S) = \frac{n}{n} = 1$$

Drawbacks: $0 \leq P(A) \leq 1$

- I) If the cases are not equally likely + II) Total No. of cases are not finite.
- #### II) Statistical or Frequency Approach:

We repeat a random experiment a very large no. of times say N (large $N \rightarrow \infty$), if out of N such repetition, an event A is being favoured m times, then the

$$P(A) = \lim_{N \rightarrow \infty} \frac{m}{N}$$

Ex. How to find $P(H)$ in a biased coin

N	10	20	30	40	50	60	...
m	m_1	m_2	m_3	m_4	m_5	m_6	...
m/N	$\frac{m_1}{10}$	$\frac{m_2}{20}$	$\frac{m_3}{30}$	$\frac{m_4}{40}$	$\frac{m_5}{50}$	$\frac{m_6}{60}$	...

→ This ratio will get stabilized.

III) Subjective Approach: Based upon one's belief

Ex. i) What is the prob that a particular train will be one hour late

ii) What is the prob that one can see a white crow fly in the sky

iii) What is the prob that my friend will be at home.

IV) Axiomatic or Set theoretic or Modern Approach

Let $\mathcal{A}$ be the σ -field defined on a sample space S of a random experiment E , the prob of an event A is a set function satisfies the following axioms:

i) For any event $A \in \mathcal{A}$
 $P(A) \geq 0$ (axiom of positivity)

ii) $P(S) = 1$ (axiom of certainty)

iii) For the mutually exclusive/mutually disjoint events $A_1, A_2, \dots, A_n \in \mathcal{A}$

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$$

$$= P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) \quad (\text{axiom of finite additivity})$$

iv) In general for any no. of m. exc./m.d. events $A_1, A_2, \dots \in \mathcal{A}$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \quad (\text{axiom of } \sigma \text{ additivity})$$

Ex. 1) $P(\phi) = 0$

$$S = S \cup \phi$$

$$P(S) = P(S \cup \phi) = P(S) + P(\phi) \quad (S + \phi \text{ are m. exc. events})$$

$$\Rightarrow P(\phi) = 0 \quad \text{by axiom (i)}$$

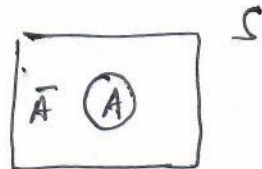
ii) $P(A) = 1 - P(\bar{A})$

$$S = A \cup \bar{A}$$

$$P(S) = P(A \cup \bar{A})$$

$$= P(A) + P(\bar{A}) \quad (\text{axiom iii})$$

$$\Rightarrow P(A) = 1 - P(\bar{A}) \quad ; \quad P(S) = 1 \quad (\text{axiom ii})$$



iii) For any two events A and B

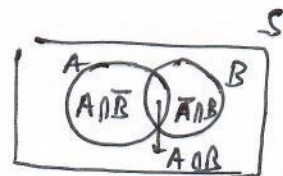
$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

Proof. $A = (A \cap \bar{B}) \cup (A \cap B)$

$$P(A) = P[(A \cap \bar{B}) \cup (A \cap B)]$$

$$= P(A \cap \bar{B}) + P(A \cap B) \quad (\text{axiom iii})$$

$$\Rightarrow \boxed{P(A \cap \bar{B}) = P(A) - P(A \cap B)}$$



Similarly $P(\bar{A} \cap B) = P(B) - P(A \cap B)$

$$\text{or } B = (\bar{A} \cap B) \cup (A \cap B)$$

$$P(B) = P(\bar{A} \cap B) + P(A \cap B)$$

$$\boxed{P(\bar{A} \cap B) = P(B) - P(A \cap B)}$$

(iv) $P[(A \cap \bar{B}) \cup (\bar{A} \cap B)] = P(A) + P(B) - 2P(A \cap B)$

As $(A \cap \bar{B})$ and $(\bar{A} \cap B)$ are m.d. events

$$P[(A \cap \bar{B}) \cup (\bar{A} \cap B)] = P(A \cap \bar{B}) + P(\bar{A} \cap B)$$

$$= P(A) - P(A \cap B) + P(B) - P(A \cap B)$$

$$= P(A) + P(B) - 2P(A \cap B)$$

* (iv) indicates the prob. of either event A occurs or event B occurs but they do not occur together.

Law of addition of any two events

For any two events A and B

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

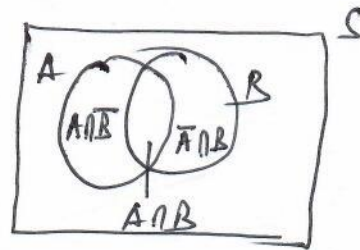
Proof: $A \cup B = A \cup (\bar{A} \cap B)$

$$P(A \cup B) = P[A \cup (\bar{A} \cap B)]$$

$$= P(A) + P(\bar{A} \cap B) \quad (\text{axiom iii})$$

$$= P(A) + P(B) - P(A \cap B)$$

Hence Proved



General Law of addition of probabilities

For any n events $A_1, A_2, \dots, A_n$

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j)$$

$$+ \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k) + \dots + (-1)^{n-1} P\left(\bigcap_{i=1}^n A_i\right)$$

Ex. For $n=3$

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$$

Boole's Inequality

Statement: For any n events $A_1, A_2, \dots, A_n$

We have (a) $P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - (n-1)$

(b) $P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i)$

Conditional Prob

Ex. H: Height W: Weight

Height & Weight of 15 students

$\overline{H}_1$	$\overline{H}_2$	H_3	$\overline{H}_4$	H_5	$\overline{H}_6$	$\overline{H}_7$	$\overline{H}_8$	H_9	$\overline{H}_{10}$	H_{11}	$\overline{H}_{12}$	H_{13}	H_{14}	H_{15}
$\underline{W}_1$	$\underline{W}_2$	$\underline{W}_3$	$\underline{W}_4$	$\underline{W}_5$	$\underline{W}_6$	$\underline{W}_7$	$\underline{W}_8$	$\underline{W}_9$	$\underline{W}_{10}$	$\underline{W}_{11}$	$\underline{W}_{12}$	$\underline{W}_{13}$	$\underline{W}_{14}$	$\underline{W}_{15}$

Unconditional Prob $P(H_7) = \frac{8}{15}$

Conditional Prob $P(H_7|W_7) = \frac{5}{89}$

Conditional Prob of the event A given B is written as

$$P(A|B)$$

Product rule for any two events A and B

$$P(A \cap B) = P(A) P(B|A) \text{ --- } (*) ; P(A) > 0$$

$$P(A \cap B) = P(B) P(A|B) \text{ --- } (**); P(B) > 0$$

Proof: $P(A \cap B) = \frac{N_{AB}}{N}$

$$= \frac{N_{AB}}{N_A} \cdot \frac{N_A}{N} = P(B|A) P(A)$$

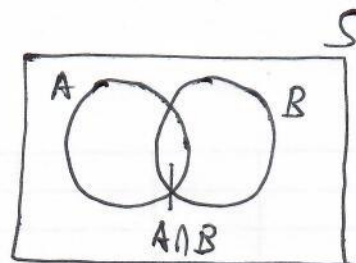
$$= P(A) P(B|A); (* \text{ proved})$$

Similarly

$$P(A \cap B) = \frac{N_{AB}}{N} = \frac{N_{AB}}{N_B} \cdot \frac{N_B}{N}$$

$$= P(A|B) P(B)$$

$$= P(B) \cdot P(A|B); (** \text{ proved})$$



$$A \rightarrow N_A \quad N_{AB} \rightarrow A \cap B$$

$$B \rightarrow N_B \quad S \rightarrow N$$

$$P(A) = \frac{N_A}{N}, \quad P(B) = \frac{N_B}{N}$$

$$P(A \cap B) = \frac{N_{AB}}{N}$$

$$P(A|B) = \frac{N_{AB}}{N_B}$$

$$P(B|A) = \frac{N_{AB}}{N_A}$$

General Product Rule for any n events

For any events $A_1, A_2, \dots, A_n$

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P\left(\bigcap_{i=1}^n A_i\right) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \\ \dots P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1})$$

Independence of events

Event A is said to be independent of the event B

if $P(A|B) = P(A)$

Similarly the event B is said to be independent of the event A if $P(B|A) = P(B)$

For the independent events A and B

$$P(A \cap B) = P(A) \cdot P(B)$$

Pairwise Independent

The events $A_1, A_2, \dots, A_n$ are said to be the pairwise independent if $P(A_i \cap A_j) = P(A_i)P(A_j); \forall i \neq j = 1, 2, \dots, n$

Mutual Independent

The events $A_1, A_2, \dots, A_n$ are said to be the mutually independent iff $P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2) \cdot \dots \cdot P(A_n)$

$$\Rightarrow P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i)$$

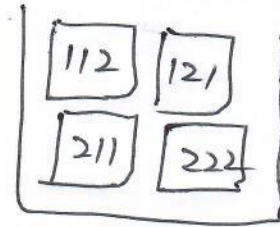
Pairwise independence does not imply mutual independence

Ex. An urn contains four tickets with numbers 112, 121, 211, 222 and one ticket is drawn at random. Let A_i ($i=1, 2, 3$) be the event that i th digit of the number of the ticket drawn is 1. Discuss the independence of the events A_1, A_2 and A_3 .

We have $P(A_1) = \frac{2}{4} = \frac{1}{2}$

$$P(A_2) = \frac{2}{4} = \frac{1}{2}$$

$$P(A_3) = \frac{2}{4} = \frac{1}{2}$$



Similarly $P(A_1 \cap A_2) = \frac{1}{4}$, $P(A_1 \cap A_3) = \frac{1}{4}$, $P(A_2 \cap A_3) = \frac{1}{4}$

$$P(A_1 \cap A_2) = P(A_1)P(A_2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Similarly $P(A_1 \cap A_3) = P(A_2 \cap A_3) = \frac{1}{4}$

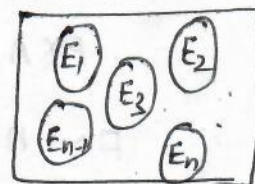
Events are pairwise independent.

But $P(A_1 \cap A_2 \cap A_3) = \frac{0}{4} = 0 \neq P(A_1) \cdot P(A_2) \cdot P(A_3)$

Hence the events are not mutually independent.

Bayes Theorem: If $E_1, E_2, \dots, E_n$ are mutually disjoint events with $P(E_i) \neq 0$ ($i=1, 2, \dots, n$) then for any arbitrary event A which is a subset of $\bigcup_{i=1}^n E_i$ such that $P(A) > 0$, we have

$$P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum_{i=1}^n P(E_i)P(A|E_i)}, \quad i=1, 2, \dots, n$$



Proof: Since $A \subset \bigcup_{i=1}^n E_i$, we have

$$A = A \cap \left(\bigcup_{i=1}^n E_i \right) = \bigcup_{i=1}^n (A \cap E_i) \quad (\text{By distributive law})$$

Since $A \cap E_i \subset E_i$ ($i=1, 2, \dots, n$) are mutually disjoint events, we have by addition theorem of prob (Axiom 3 of probability)

$$P(A) = P\left(\bigcup_{i=1}^n (A \cap E_i)\right) = \sum_{i=1}^n P(A \cap E_i) = \sum_{i=1}^n P(E_i)P(A|E_i)$$

$$P(E_i|A) = \frac{P(A \cap E_i)}{P(A)} = \frac{P(E_i)P(A|E_i)}{\sum_{i=1}^n P(E_i)P(A|E_i)}$$

Remark: (i) $P(E_i)$ ($i=1, 2, \dots, n$) are known as a priori probabilities
 (ii) $P(A|E_i)$ ($i=1, 2, \dots, n$) are known as likelihoods
 (iii) $P(E_i|A)$ ($i=1, 2, \dots, n$) are called posterior probabilities.